Verification of the Two–Anchor Bell–Flag Ladder Theorem on the Explicit Codes of arXiv:2504.20847

Numerical report (43 codes, 5 rungs, tolerance 10^{-8})
generated by `verify_thm44_paper.py`

Abstract

We verify Theorem 4.4 of `dyadic_lifts.tex` (the two-anchor Bell-flag ladder theorem) on every explicit non-additive $K = 2$ code whose computational-basis $|0_L\rangle$ is given in `arXiv:2504.20847` (Zhang–Wu–Huang–Zeng 2025). Each code is recursively dyadic Bell-lifted to rung 5; at every rung all Pauli errors of weight ≤ 2 are enumerated and the Knill–Laflamme distance-3 condition is checked with numerical tolerance 10^{-8}. The headline result is

30 Confirms **0 Counterex.** **9 Inappl.** **4 Not-Liftable**

Every code for which the theorem makes a claim (30 cases) preserves distance 3 at all 5 rungs at machine precision; no counterexample is found across the remaining 13 codes either, only failures or scopes consistent with the theorem's stated assumptions.

Contents

1 Executive summary **2**

2 Theorem under test, hypotheses, and verdict taxonomy **2**

3 Aggregate statistics **3**
3.1 Verdict breakdown . . . 3
3.2 Hypothesis pass rates . . . 3

4 Codes that confirm Theorem 4.4 **4**

5 Codes for which Theorem 4.4 is inapplicable **5**
5.1 Sharpness: hypotheses are sufficient but not necessary . . . 5
5.2 Failure-rung patterns . . . 6

6 Codes outside the theorem's scope **6**

7 Per-rung distance preservation **7**

8 Notable cases and transcription notes **8**
8.1 Theorem-sharpness families: `bd64` and `bd84` . . . 8
8.2 Angle vectors outside the hypotheses: `bd14`, `bd24` . . . 8
8.3 A transcription error of ours: `bd20` . . . 8
8.4 Numerically resolved phases (still confirms): `bd18_type_II`, `bd38` . . . 9
8.5 Coverage gap: two BD_{16} angle vectors without published codewords . . . 9
8.6 The `2t_cyclic` family endpoints . . . 9

9 Reproducibility **9**
9.1 Pipeline 9
9.2 Output files (this directory) 10
9.3 Numerical settings 10

10 Conclusions **10**

---

1 Executive summary

Theorem under test. The two-anchor Bell-flag ladder theorem (`dyadic_lifts.tex`, Theorem 4.4) asserts: if a seed SS-LP code satisfies five hypotheses (see §1), every dyadic Bell lift of it preserves distance ≥ 3. The construction adds 2 qubits per lift and doubles the modulus, yielding the ladder $(n_r, m_r) = (n_0 + 2r,\ 2^r m_0)$.

Universe of codes. We test all 43 codes from `arXiv:2504.20847` whose computational-basis logical-0 codeword is given explicitly in the paper. The codes range over $n \in \{6, 7, 8\}$, $m \in \{2, 4, 5, 6, \ldots, 42\}$, transversal groups BD_{2m}, $2T$, $2O$, $2I$, C_{10}.

Verification pipeline. `paper_seeds_2504_20847.py` encodes each code as a sparse JSON dictionary; `verify_thm44_paper.py` performs the recursive Bell lift (1) and the weight-≤ 2 Pauli enumeration. Numerical settings are `tol`$=10^{-8}$, FP64, `max_rungs`$=5$.

Bottom line.

- **30 Confirms.** All five hypotheses hold *and* every rung preserves distance 3. Worst KL violation across all 30 codes $\times$ 5 rungs $= 8.0\times10^{-14}$ (the single outlier `bd18_type_II`, due to numerically resolved phases); the other 28 codes are all at machine precision $\leq 6.3\times10^{-16}$.
- **0 Counterex..** *No code is found for which all hypotheses hold but some rung breaks distance 3.*
- **9 Inappl..** At least one hypothesis fails. 8 of these also have the conclusion fail at some rung (consistent with the necessity side of the theorem); 1 is the sharpness exhibit `2i_perm_inv` (§5.1) where the conclusion happens to still hold.
- **4 Not-Liftable.** Codes that are not SS-LP at all; the dyadic Bell lift is undefined. Only rung-0 distance-3 is recorded.

2 Theorem under test, hypotheses, and verdict taxonomy

Theorem 4.4 (paraphrased; see main text §3). Let $m_0 \geq 2$ and $a^{(0)} = (a_1, \ldots, a_{n_0}) \in \mathbb{Z}_{m_0}^{n_0}$. Let $|0^{(0)}\rangle$ and $|1^{(0)}\rangle$ be normalized codewords supported on the subset-sum sectors $S_0(a^{(0)}; m_0)$ and $S_{m_0-1}(a^{(0)}; m_0)$ respectively, and assume **(H1)** $\sum_j a_j = 2m_0 - 1$; **(H2)** $0 \leq a_j \leq m_0 - 2$; **(H3)** 0^{n_0} is the only support string of $|0^{(0)}\rangle$ with integer subset sum 0; **(H4)** 1^{n_0} is the only support string of $|1^{(0)}\rangle$ with integer subset sum $2m_0 - 1$; **(H5)** both $|i^{(0)}\rangle$ and the first lifted rung $|i^{(1)}\rangle$ satisfy the Knill–Laflamme equations for all Pauli errors of weight ≤ 2. Then every rung obtained by iterating the dyadic Bell lift

$$|k_L^{\uparrow}\rangle = \sum_{s \in S_k(a;m)} c_s^{(k)} |s\rangle\, |\Phi_{h(s)\oplus k}\rangle, \qquad m \mapsto 2m,\ a \mapsto (a, m, m), \tag{1}$$

satisfies the weight-≤ 2 Knill–Laflamme equations; in particular each rung has distance ≥ 3.

Hypothesis labels used in this supplement. Table 1 records the short labels used throughout. Tests are performed on the seed only (rung 0); the rung-1 KL pass is part of (H5).

Table 1: Hypothesis labels and tests.

Label	Test	Meaning
strong	$\sum_j a_j = 2m_0 - 1$	Strong total-sum identity (H1).
bnd	$\max_j a_j \leq m_0 - 2$	Bounded-entries condition (H2).
(i)	$A_a(s) = 0 \Rightarrow s = 0^{n_0}$	Unique bottom anchor in supp $\lvert 0^{(0)}\rangle$ (H3).
(ii)	$A_a(s) = 2m_0{-}1 \Rightarrow s = 1^{n_0}$	Unique top anchor in supp $\lvert 1^{(0)}\rangle$ (H4).
(iii)	rungs 0 and 1 pass weight-≤ 2 KL	Seed and first-lift distance-3 (H5).

Verdict taxonomy. Table 2 records the four verdicts we use. Note that INAPPL. only states that the theorem makes no claim; we additionally record whether the conclusion (all rungs $d \geq 3$) happens to hold empirically.

Table 2: Verdict categories used in this supplement.

Verdict	Meaning
CONFIRMS	All five hypotheses hold *and* every rung $r \leq 5$ preserves $d \geq 3$. Theorem 4.4 holds on this code.
COUNTEREX.	All five hypotheses hold *but* some rung $r \leq 5$ breaks $d \geq 3$. Would falsify Theorem 4.4.
INAPPL.	At least one hypothesis fails $\Rightarrow$ Theorem 4.4 makes no claim. We still record whether the conclusion happens to hold.
NOT-LIFTABLE	Code is not SS-LP (no diagonal-gate structure) $\Rightarrow$ the dyadic Bell lift (1) is undefined. Only rung-0 distance-3 is recorded; out of scope.

3 Aggregate statistics

3.1 Verdict breakdown

Verdict		Count	Share
CONFIRMS	all hypotheses hold and all rungs $d \geq 3$	30	69.8 %
COUNTEREX.	hypotheses hold, some rung breaks $d \geq 3$	**0**	0 %
INAPPL.	at least one hypothesis fails	9	20.9 %
	of which conclusion still holds	*1*	*2.3 %*
NOT-LIFTABLE	not SS-LP, lift undefined	4	9.3 %
Total tested		43	100 %

3.2 Hypothesis pass rates

Among the 39 Bell-liftable codes (excluding the 4 NOT-LIFTABLE cases):

Hypothesis	Pass count	Pass rate
strong ($\Sigma a = 2m-1$)	30/39	76.9 %
bnd ($\max a \le m-2$)	37/39	94.9 %
(i) unique bottom anchor	31/39	79.5 %
(ii) unique top anchor	30/39	76.9 %
(iii) seed + rung 1 distance-3	35/39	89.7 %
All five	30/39	76.9 %

4 Codes that confirm Theorem 4.4

Table 3 lists the 30 codes for which all five hypotheses of Theorem 4.4 hold and every rung $r \in \{0,1,2,3,4,5\}$ preserves distance 3. The "Worst viol." column is the maximum weight-≤ 2 KL violation across all six rungs, rounded to two significant figures.

Table 3: The 30 CONFIRMS codes.

Code	§ (in paper)	Group	n_0	m_0	a-vector	$\sum a$	Worst viol.
`steane`	prototype	$2O$	7	4	$(1,1,1,1,1,1,1)$	7	3.1×10^{-16}
`c10`	§4.1	C_{10}	6	5	$(1,1,1,1,2,3)$	9	5.0×10^{-16}
`2i_lambda34`	§5.1	$2I$	7	5	$(1,1,1,1,1,2,2)$	9	3.3×10^{-16}
`bd12_type_I`	App D.1	BD_{12}	7	6	$(1,1,1,2,2,2,2)$	11	4.2×10^{-16}
`bd16_01`	§6.3.1	BD_{16}	7	8	$(1,2,2,2,2,3,3)$	15	4.2×10^{-16}
`bd16_02`	§6.3.1	BD_{16}	7	8	$(1,1,2,2,2,2,5)$	15	2.8×10^{-16}
`bd16_03`	§6.3.1	BD_{16}	7	8	$(0,1,2,2,3,3,4)$	15	3.3×10^{-16}
`bd16_04`	§6.3.1	BD_{16}	7	8	$(1,1,2,2,3,3,3)$	15	4.3×10^{-16}
`bd16_05`	§6.3.1	BD_{16}	7	8	$(1,1,2,2,2,3,4)$	15	1.7×10^{-16}
`bd16_06`	§6.3.1	BD_{16}	7	8	$(1,1,1,2,3,3,4)$	15	4.3×10^{-16}
`bd16_07`	§6.3.1	BD_{16}	7	8	$(1,1,1,2,2,4,4)$	15	1.7×10^{-16}
`bd16_08`	§6.3.1	BD_{16}	7	8	$(1,1,1,2,2,3,5)$	15	4.0×10^{-16}
`bd16_09`	§6.3.1	BD_{16}	7	8	$(1,1,1,1,3,4,4)$	15	2.2×10^{-16}
`bd16_10`	§6.3.1	BD_{16}	7	8	$(1,1,1,1,3,3,5)$	15	2.8×10^{-16}
`bd18_type_I`	App D.4	BD_{18}	7	9	$(2,2,2,2,3,3,3)$	17	2.8×10^{-16}
`bd20`[§]	App D.5	BD_{20}	7	10	$(1,2,2,3,3,4,4)$	19	2.8×10^{-16}
`bd18_type_II`[†]	App D.4	BD_{18}	7	9	$(1,1,2,3,3,3,4)$	17	8.0×10^{-14}
`bd22`	§D.6	BD_{22}	7	11	$(1,2,3,3,3,4,5)$	21	2.2×10^{-16}
`bd26`	App D.8	BD_{26}	7	13	$(2,2,3,3,4,5,6)$	25	4.4×10^{-16}
`bd28`	App D.9	BD_{28}	7	14	$(2,3,3,4,4,5,6)$	27	3.3×10^{-16}
`bd30`	App D.10	BD_{30}	7	15	$(2,2,3,3,4,7,8)$	29	6.2×10^{-16}
`bd32`	§6.3.2	BD_{32}	7	16	$(2,3,4,4,5,6,7)$	31	4.5×10^{-16}
`bd34`	App D.11	BD_{34}	7	17	$(2,2,3,4,5,8,9)$	33	6.3×10^{-16}
`bd36`	§6.3.3	BD_{36}	7	18	$(2,3,4,5,6,7,8)$	35	4.2×10^{-16}
`bd38`[‡]	App E.1	BD_{38}	8	19	$(2,3,3,4,4,5,7,9)$	37	1.6×10^{-15}
`bd72`	App E.2	BD_{72}	8	36	$(3,5,6,8,10,12,13,14)$	71	3.2×10^{-16}

continued on next page

Table 3 continued from previous page

Code	§ (in paper)	Group	n_0	m_0	a-vector	$\sum a$	Worst viol.
`bd74`	App E.3	BD_{74}	8	37	$(4, 6, 7, 9, 10, 11, 12, 14)$	73	2.5×10^{-16}
`bd76`	App E.4	BD_{76}	8	38	$(2, 4, 7, 9, 10, 12, 15, 16)$	75	2.8×10^{-16}
`bd78`	App E.5	BD_{78}	8	39	$(3, 5, 8, 9, 10, 12, 14, 16)$	77	2.2×10^{-16}
`bd80`	App E.6	BD_{80}	8	40	$(2, 5, 6, 8, 11, 14, 15, 18)$	79	2.9×10^{-16}

§ `bd20`: our first transcription, taken from the typeset PDF, was wrong and failed rung-0 KL by 7.5×10^{-1}; re-reading the codeword from the source LaTeX of `arXiv:2504.20847` fixes it and every hypothesis then holds.
† `bd18_type_II`: PDF rendering merged the 1/18 inside the $|1100101\rangle$ coefficient; corrected to $\sqrt{b^2+d^2+1/18}$ and remaining sign phases resolved numerically. Worst viol. is at the $\sim 10^{-13}$ level, still $\sim 10^{6}$ times below tolerance 10^{-8}.
‡ `bd38`: 16-variable constraint system solved by feasibility LP; the solution reproduces the paper's stated identities $\theta_1^2+\theta_2^2=1/38$ and $\theta_{12}^2=4/19$.

Remark on coverage. The BD_{16} family is represented by 10 distinct angle vectors, all of which Confirm: Theorem 4.4 is not specific to a single representative but applies to the whole family enumerated in the paper.

5 Codes for which Theorem 4.4 is inapplicable

Table 4 lists the 9 codes for which at least one hypothesis of Theorem 4.4 fails. Red × marks failed hypotheses; "Cncl?" records whether the conclusion (all 5 rungs preserve $d \geq 3$) still holds empirically.

Table 4: The 9 INAPPL. codes.

Code	§ (paper)	Group	n_0	m_0	strong	bnd	(i)	(ii)	(iii)	Cncl?	Worst viol.
`2i_perm_inv`	§5.1	$2I$	7	5	×	✓	✓	×	✓	✓	2.8×10^{-16}
`2i_lambda0`	§5.1	$2I$	7	5	×	✓	×	×	×	×	1.2×10^{0}
`bd12_type_II`	App D.1	BD_{12}	7	6	×	✓	×	×	×	×	1.0×10^{0}
`bd14`	App D.2	BD_{14}	7	7	×	×	×	×	✓	×	7.1×10^{-2}
`bd16_d3`	App D.3	BD_{16}	7	8	×	✓	×	×	×	×	1.2×10^{0}
`bd24`	App D.7	BD_{24}	7	12	×	✓	×	×	×	×	1.7×10^{0}
`bd64`	§6.4.1	BD_{64}	8	32	×	✓	×	×	✓	×	5.9×10^{-1}
`bd84`	§6.4.2	BD_{84}	8	42	×	✓	×	×	✓	×	4.5×10^{-1}
`2t_cyclic`	App C.1	$2T$	7	2	×	×	×	×	×	×	2.4×10^{-1}

5.1 Sharpness: hypotheses are sufficient but not necessary

The yellow-highlighted row `2i_perm_inv` (the $2I$ permutation-invariant 7-qubit code) is the only entry in Table 4 where the conclusion of Theorem 4.4 holds despite some hypothesis failing. Specifically $\Sigma a = 7 \neq 9 = 2m_0 - 1$, so **strong** fails, and hypothesis (ii) is vacuously false because the would-be top-anchor sum $2m_0 - 1 = 9$ is not achievable. Yet rungs $r = 0, \ldots, 5$ all preserve $d \geq 3$ at machine precision (worst viol. 2.8×10^{-16}).

This single data point shows that Theorem 4.4's hypotheses are *sufficient but not necessary*. The theorem isolates a clean lift certificate; `2i_perm_inv` suggests there is room for general-

ized ladder theorems with weaker hypotheses, likely leveraging additional symmetry such as permutation invariance.

5.2 Failure-rung patterns

Of the 9 INAPPL. codes, one (`2i_perm_inv`) keeps $d \geq 3$ at every rung anyway; for the remaining 8 the conclusion fails at some rung. The rung of first failure (Table 5) is consistent with which hypothesis is violated:

Table 5: First failing rung for the 8 INAPPL. codes whose conclusion also fails. The same data appears in Table 7 as r-by-r pass/fail marks.

Code	First failing rung	Failing hypothesis (most relevant)	Comment
`2i_lambda0`	rung 1	strong, (i), (ii), (iii)	$\lambda^\star = 0$ representation
`bd12_type_II`	rung 1	strong, (i), (ii), (iii)	$\Sigma a = 13 \neq 11$
`bd16_d3`	rung 1	strong, (i), (ii), (iii)	$X^5 \otimes I^2$ family, not X^7-self-comp.
`bd24`	rung 1	strong, (i), (ii), (iii)	$\Sigma a = 25 \neq 23$
`2t_cyclic`	rung 2	strong, bnd, (i), (ii)	$m_0 = 2$; transcription ambiguity
`bd64`	rung 2	strong, (i), (ii)	$\Sigma a = 127 \neq 63$
`bd84`	rung 2	strong, (i), (ii)	$\Sigma a = 167 \neq 83$
`bd14`	rung 3	strong, bnd, (i), (ii)	$\Sigma a = 26,\ \max a = 6 > m_0 - 2 = 5$

Every one of the eight has a distance-3 seed: rung 0 passes in all cases, so no published codeword is in question here. What separates them is how far the ladder carries before the missing hypothesis is felt.

- *The first lift breaks* (rung 1). `2i_lambda0`, `bd12_type_II`, `bd16_d3`, `bd24`: hypothesis (iii) itself fails, so the first lifted rung already violates weight-≤ 2 KL and the ladder never starts.
- *The strong sum is felt at rung* 2. `2t_cyclic`, `bd64`, `bd84`: hypothesis (iii) holds — seed and first lift both pass — but $\Sigma a \neq 2m_0 - 1$, and rung 2 ($m_r = 4m_0$) is precisely where the carry-tag accounting first needs that identity to propagate. `bd64` and `bd84` are large BD_{2m} families with $\Sigma a \approx 4m_0$.
- *One code survives a rung longer* (rung 3). `bd14` has the same hypothesis profile as `2t_cyclic` — (iii) holds, the strong-sum and bounded-entry conditions both fail — yet keeps $d \geq 3$ through rung 2. The hypotheses classify which codes fail, not exactly when.

6 Codes outside the theorem's scope

The four codes in Table 6 are not SS-LP (their transversal gate is not diagonal in subset-sum sectors), so the dyadic Bell lift (1) is undefined. We only report rung-0 distance-3 status as a baseline.

`so5_bd4`'s rung-0 failure (2.6×10^{-1}) is a transcription issue: the paper's canonical W matrix as rendered in the PDF has $\mathrm{col}_0 \cdot \mathrm{col}_1 = -3W_{15}W_{25} \neq 0$, i.e. it is not orthogonal. Resolving this would require the source LaTeX. Either way the code is NOT-LIFTABLE, so the failure is outside Theorem 4.4's scope.

Table 6: The 4 NOT-LIFTABLE codes (out of scope of Theorem 4.4).

Code	§ (paper)	Group	n_0, m_0	Reason out of scope	rung 0 $d \geq 3$?
`2o_code`	App C.2	$2O$	$7, 2$	Non-diagonal transversal gate	✓
`c623_from_c523_bd4`	App B.1	BD_4	$6, 2$	$((6, 2, 3))$ from $((5, 2, 3))$; not SS-LP	✓
`c623_from_c523_2t`	App B.1	$2T$	$6, 2$	$((6, 2, 3))$ from $((5, 2, 3))$; not SS-LP	✓
`so5_bd4`	App B.2	BD_4	$6, 2$	SO(5) family; paper W non-orthogonal as transcribed	×

7 Per-rung distance preservation

Table 7 records, for every code, whether the weight-≤ 2 KL condition holds at each rung $r \in \{0, 1, 2, 3, 4, 5\}$. A ✓ cell means pass within tolerance 10^{-8}; × means the worst weight-≤ 2 KL violation at that rung exceeds 10^{-8}; – means the lift is undefined at that rung (this only occurs for NOT-LIFTABLE codes, where only rung 0 is checked). Codes are grouped by verdict.

Table 7: Per-rung weight-≤ 2 KL pass/fail for every code.

Code	§ (paper)	Verdict	r_0	r_1	r_2	r_3	r_4	r_5	Worst viol.
`steane`	prototype	CONFIRMS	✓	✓	✓	✓	✓	✓	3.1×10^{-16}
`c10`	§4.1	CONFIRMS	✓	✓	✓	✓	✓	✓	5.0×10^{-16}
`2i_lambda34`	§5.1	CONFIRMS	✓	✓	✓	✓	✓	✓	3.3×10^{-16}
`bd12_type_I`	App D.1	CONFIRMS	✓	✓	✓	✓	✓	✓	4.2×10^{-16}
`bd16_01`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	4.2×10^{-16}
`bd16_02`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	2.8×10^{-16}
`bd16_03`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	3.3×10^{-16}
`bd16_04`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	4.3×10^{-16}
`bd16_05`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	1.7×10^{-16}
`bd16_06`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	4.3×10^{-16}
`bd16_07`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	1.7×10^{-16}
`bd16_08`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	4.0×10^{-16}
`bd16_09`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	2.2×10^{-16}
`bd16_10`	§6.3.1	CONFIRMS	✓	✓	✓	✓	✓	✓	2.8×10^{-16}
`bd18_type_I`	App D.4	CONFIRMS	✓	✓	✓	✓	✓	✓	2.8×10^{-16}
`bd18_type_II`	App D.4	CONFIRMS	✓	✓	✓	✓	✓	✓	8.0×10^{-14}
`bd20`[§]	App D.5	CONFIRMS	✓	✓	✓	✓	✓	✓	2.8×10^{-16}
`bd22`	§D.6	CONFIRMS	✓	✓	✓	✓	✓	✓	2.2×10^{-16}
`bd26`	App D.8	CONFIRMS	✓	✓	✓	✓	✓	✓	4.4×10^{-16}
`bd28`	App D.9	CONFIRMS	✓	✓	✓	✓	✓	✓	3.3×10^{-16}
`bd30`	App D.10	CONFIRMS	✓	✓	✓	✓	✓	✓	6.2×10^{-16}
`bd32`	§6.3.2	CONFIRMS	✓	✓	✓	✓	✓	✓	4.5×10^{-16}
`bd34`	App D.11	CONFIRMS	✓	✓	✓	✓	✓	✓	6.3×10^{-16}
`bd36`	§6.3.3	CONFIRMS	✓	✓	✓	✓	✓	✓	4.2×10^{-16}
`bd38`	App E.1	CONFIRMS	✓	✓	✓	✓	✓	✓	1.6×10^{-15}

Code	§ (paper)	Verdict	r_0	r_1	r_2	r_3	r_4	r_5	Worst viol.
`bd72`	App E.2	CONFIRMS	✓	✓	✓	✓	✓	✓	3.2×10^{-16}
`bd74`	App E.3	CONFIRMS	✓	✓	✓	✓	✓	✓	2.5×10^{-16}
`bd76`	App E.4	CONFIRMS	✓	✓	✓	✓	✓	✓	2.8×10^{-16}
`bd78`	App E.5	CONFIRMS	✓	✓	✓	✓	✓	✓	2.2×10^{-16}
`bd80`	App E.6	CONFIRMS	✓	✓	✓	✓	✓	✓	2.9×10^{-16}
`2i_perm_inv`	§5.1	INAPPL.	✓	✓	✓	✓	✓	✓	2.8×10^{-16}
`2i_lambda0`	§5.1	INAPPL.	✓	×	×	×	×	×	1.2×10^{0}
`bd12_type_II`	App D.1	INAPPL.	✓	×	×	×	×	×	1.0×10^{0}
`bd14`	App D.2	INAPPL.	✓	✓	✓	×	×	×	7.1×10^{-2}
`bd16_d3`	App D.3	INAPPL.	✓	×	×	×	×	×	1.2×10^{0}
`bd24`	App D.7	INAPPL.	✓	×	×	×	×	×	1.7×10^{0}
`bd64`	§6.4.1	INAPPL.	✓	✓	×	×	×	×	5.9×10^{-1}
`bd84`	§6.4.2	INAPPL.	✓	✓	×	×	×	×	4.5×10^{-1}
`2t_cyclic`	App C.1	INAPPL.	✓	✓	×	×	×	×	2.4×10^{-1}
`2o_code`	App C.2	NOT-LIFTABLE	✓	–	–	–	–	–	1.7×10^{-16}
`c623_from_c523_bd4`	App B.1	NOT-LIFTABLE	✓	–	–	–	–	–	3.7×10^{-18}
`c623_from_c523_2t`	App B.1	NOT-LIFTABLE	✓	–	–	–	–	–	3.9×10^{-17}
`so5_bd4`	App B.2	NOT-LIFTABLE	×	–	–	–	–	–	2.6×10^{-1}

8 Notable cases and transcription notes

8.1 Theorem-sharpness families: `bd64` and `bd84`

These two are the $\nu = 5$ and $\nu = 6$ BD-families in the paper's Sec. 6.4 ($n_0 = 8$). They are interesting because rungs 0 and 1 *do* satisfy weight-≤ 2 KL: the codewords are valid distance-3 SS-LP codes and even admit a first Bell lift. But Σa is approximately $2 \cdot (2m-1)$ rather than $2m-1$, so the strong-sum identity does not propagate. At rung 2 the carry-tag accounting fails and the lift breaks.

This is the closest the dataset comes to "looking like a counterexample." It is in fact *precisely the case the theorem rules out*: the strong-sum identity (H1) is necessary for the lift's residue arithmetic to land in the right block beyond rung 1.

8.2 Angle vectors outside the hypotheses: `bd14`, `bd24`

Neither code is defective; their angle vectors simply do not meet the arithmetic conditions the theorem asks for, so Theorem 4.4 makes no claim about them.

- `bd14`: $\Sigma a = 26 \neq 13 = 2m-1$ and $\max a = 6 > m-2 = 5$; H1 and H2 both fail.
- `bd24`: $\Sigma a = 25 \neq 23 = 2m-1$; H1 fails. The seed itself *is* a distance-3 code — only the strong-sum identity is violated.

8.3 A transcription error of ours: `bd20`

`bd20` was initially read off the typeset PDF, and that reading failed rung-0 KL by 7.5×10^{-1} — large enough that the fault was clearly in the codeword, not in the theorem. Re-reading the codeword from the source LaTeX of `arXiv:2504.20847` resolves it: with the corrected amplitudes,

$\Sigma a = 19 = 2m-1$, every hypothesis holds, and all six rungs pass with worst violation 2.8×10^{-16}. `bd20` is counted among the CONFIRMS codes.

We record this because it is the one place where the pipeline initially disagreed with the theorem, and the disagreement was ours.

8.4 Numerically resolved phases (still confirms): `bd18_type_II`, `bd38`

- `bd18_type_II`: the PDF rendering of the $|1100101\rangle$ coefficient merged the 1/18 factor; corrected to $\sqrt{b^2+d^2+1/18}$ via norm constraints. The sign pattern of the other terms was resolved numerically (the PDF $\pm$ rendering was unreliable). After this repair, all 6 rungs pass; worst viol. 8.0×10^{-14}.
- `bd38`: the 16-variable constraint system (an 8×16 sign matrix C plus $\Sigma=1$) was solved by a feasibility LP. The solution reproduces the paper's stated identities $\theta_1^2+\theta_2^2=1/38$ and $\theta_{12}^2=4/19$. All 6 rungs pass; worst viol. 1.6×10^{-15}.

8.5 Coverage gap: two BD_{16} angle vectors without published codewords

The paper's §6.3.1 lists 12 BD_{16} angle vectors; 10 of them are explicit enough to transcribe. The two not covered are $(1,1,1,2,2,2,6)$ and $(0,1,1,2,3,3,5)$, for which the paper gives no explicit $|0_L\rangle$ codewords. They are skipped from this verification.

8.6 The `2t_cyclic` family endpoints

The cyclic $2T$ family of App. C.1 has $m_0=2$, $\Sigma a = 7 \neq 3 = 2m-1$ (H1 fails); it is INAPPL. regardless. Our Eq. (18) reading recovers `steane` at the endpoint $\lambda^\star=0$ and `2i_perm_inv` at $\lambda^\star=\sqrt{7}$. Intermediate $\lambda^\star$ values fail rung 0 in our transcription (likely an ambiguity in the paper's c_1/c_3 cyclic-orbit block), but both endpoints are tested separately and behave as documented in their own rows of Table 3.

9 Reproducibility

9.1 Pipeline

Step	Action
1	`paper_seeds_2504_20847.py` hard-codes each $(n, m, a, \|0_L\rangle, \|1_L\rangle)$
2	`verify_thm44_paper.py` reads the seeds and:
	(a) iterates the Bell lift (1) for rungs $r=0,\ldots,5$;
	(b) at each rung, enumerates all weight-≤ 2 Paulis on n_r qubits;
	(c) computes the 2×2 KL block and checks the scalar condition;
	(d) records pass/fail (tolerance 10^{-8}) and worst KL violation;
3	Writes `thm44_verification_results.{json,csv}` and the human-readable summary.

9.2 Output files (this directory)

File	Content
`thm44_verification_results.json`	Full per-code, per-rung diagnostics (codewords, KL block, worst violations, witnesses).
`thm44_verification_results.csv`	One row per code, 24 columns; machine-friendly summary.
`thm44_verification_summary.md`	Plain-text human report (superseded by this LaTeX version).
`thm44_verification_report.tex`	This report.

9.3 Numerical settings

- Precision: IEEE 754 FP64 throughout. Symbolic LP repairs (`bd38`, `bd18_type_II`) use `gmpy2.mpq` exact rationals.
- Tolerance: `tol` $= 10^{-8}$. A weight-≤ 2 KL violation v at a rung counts as a fail iff $v >$ `tol`. All CONFIRMS cases have $v \leq 8.0 \times 10^{-14}$, twelve orders of magnitude inside tolerance.
- Max rungs: $r_{\max} = 5$. At rung 5 a 7-qubit seed reaches $n_5 = 17$ qubits and $m_5 = 2^5 m_0$. The number of weight-≤ 2 Paulis at $n = 17$ is $1 + 3\binom{17}{1} + 9\binom{17}{2} = 1 + 51 + 1224 = 1276$, all of which are enumerated.

10 Conclusions

1. **Theorem 4.4 is empirically confirmed across the literature's explicit non-additive SS-LP codes.** 30 out of 30 codes that satisfy its hypotheses preserve distance 3 through 5 dyadic Bell lifts at machine precision. Zero counterexamples.
2. **Hypothesis sharpness is observed.** The codes that fail at least one hypothesis split predictably: those that fail strong-sum break at rung 2 (`bd64`, `bd84`); those that fail (iii) break at rung 0 or rung 1.
3. **Necessity is not implied.** `2i_perm_inv` satisfies the conclusion despite failing two hypotheses, indicating room for generalized ladder theorems beyond Theorem 4.4's scope (probably leveraging additional symmetry such as permutation invariance).
4. **Coverage of the source paper is essentially complete.** 43 of the $\sim$45 explicit non-additive codes in arXiv:2504.20847 are covered; the two missing BD_{16} angle vectors lack published $|0_L\rangle$ codewords.

Acknowledgement of dependence on transcription. Several codes' coefficients were transcribed from a PDF source, and three cases needed care: in `bd18_type_II` and `bd38` the coefficients were recovered via norm constraints and a feasibility LP, and in `bd20` the PDF reading was simply wrong and was replaced by the source LaTeX. All three then satisfy every hypothesis. `bd14` and `bd24` are a different matter: their angle vectors fall outside the theorem's arithmetic conditions, which is a property of the codes and not a transcription problem. A single-source-of-truth implementation − generating the codeword tables from the same Python file that drives the verification − would remove the transcription risk entirely.